

Sec 6.2 – Constructing Antiderivatives Analytically

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

If $F(x)$ is an antiderivative of $f(x)$ (where $f(x)$ is equal to $F'(x)$), then $F(x) + C$ is an antiderivative of $f(x)$ for any constant C .

So, the constants don't matter.

The **indefinite integral** of a function f is all the antiderivatives of that function. If $F(x)$ is an antiderivative of $f(x)$, we use the notation

$$\int f(x) dx = F(x) + C.$$

Observations:

- This notation is reminiscent of the definite integral because the FTC tells us there is a connection between integration and antiderivatives.
- Since the indefinite integral is *all* the antiderivatives, you must include the “ $+C$ ” or else your answer is incorrect.
- An indefinite integral is a *family of functions*, but a definite integral is a *number*. While the processes are related, the meaning of the result is very different!

$$\int_a^b f(x) dx \text{ and } \int f(x) dx.$$

Finding antiderivatives analytically, which means finding the family of functions, is like taking square roots of numbers. The square root of 8 or 145 might be difficult to calculate, but the square root of 9 or 144 are pretty easy since they are perfect squares.

So, if we pick a function which is a derivative of another function, like $2x$, which is the derivative of x^2 , then it is easy to see that $x^2 + C$ is the antiderivative of $2x$.

This leads us to some rules which “undo” the derivatives:

$$\text{If } n \neq -1, \int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

$$\text{If } n = -1, \int x^{-1} dx = \int \frac{1}{x} dx = \ln |x| + C$$

Why the $|x|$ in the logarithm? $\ln(x)$ is only defined for $x > 0$, but $\frac{1}{x}$ is defined for all $x \neq 0$, so we need an antiderivative that works for $x < 0$.

And just undoing derivatives gives us:

$$\boxed{\int \cos x dx = \sin x + C} \quad \text{and} \quad \boxed{\int \sin x dx = -\cos x + C.}$$

Partner up

1. $\int k \, dx$

2. $\int 2x \, dx$

3. $\int x^2 \, dx$

4. $\int 2x^4 \, dx$

5. $\int \frac{1}{x} \, dx$

6. $\int \frac{1}{x^2} \, dx$

7. $\int e^x \, dx$

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1. $\int k \, dx = kx + C$

2. $\int 2x \, dx = x^2 + C$

3. $\int x^2 \, dx = \frac{x^3}{3} + C$

4. $\int 2x^4 \, dx = \frac{2x^5}{5} + C$

5. $\int \frac{1}{x} \, dx = \ln x + C$ for $x > 0^*$

6. $\int \frac{1}{x^2} \, dx = -\frac{1}{x} + C$

7. $\int e^x \, dx = e^x + C$

Partner up

8. $\int \cos x \, dx$

9. $\int \sin x \, dx$

10. $\int \frac{1}{\cos^2 x} \, dx$

11. $\int \frac{1}{1+x^2} \, dx$

12. $\int \frac{1}{\sqrt{1-x^2}} \, dx$

13. $\int 3^x \, dx$

14. $\int \cos(2x) \, dx$

8. $\int \cos x \, dx = \sin x + C$

9. $\int \sin x \, dx = -\cos x + C$

10. $\int \frac{1}{\cos^2 x} \, dx = \tan x + C$

11. $\int \frac{1}{1+x^2} \, dx = \arctan x + C$

12. $\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$

13. $\int 3^x \, dx = \frac{3^x}{\ln 3} + C$

14. $\int \cos(2x) \, dx = \frac{1}{2} \sin(2x) + C$

In indefinite integral notation,

$$\textcircled{1} \int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

An antiderivative of the sum (or difference) of two functions is the sum (or difference) of their antiderivatives.

$$\textcircled{2} \int cf(x) dx = c \int f(x) dx$$

An antiderivative of a constant times a function is the constant times an antiderivative of the function.

Note: while analogous to our properties for definite integrals, these properties are about families of functions rather than numbers!

$$\text{Find } \int x^2 + \frac{3}{x} - 4 \cos x \, dx = \frac{x^3}{3} + 3 \ln |x| - 4 \sin x + C$$

As we get better at finding the antiderivative functions, we can more easily use the FTC to evaluate definite integrals.

Evaluate the definite integral: $\int_1^3 4x^3 \, dx$.

Since x^4 is an antiderivative of $4x^3$, the FTC says that

$$\int_1^3 4x^3 \, dx = x^4 \Big|_1^3 = 3^4 - 1^4 = 81 - 1 = 80.$$

By the way, if you wanted to evaluate $\int_1^3 (4x^3 + 2) dx$, you would get the same answer because the “+2” terms cancel in the subtraction, which will happen with any value of C .

Partner up

Find the following integrals. Note that some are indefinite integrals and some are definite integrals - make sure to answer in the correct form!

$$\textcircled{1} \int 3\sqrt{x} - \frac{1}{1+x^2} \, dx$$

$$\textcircled{2} \int_1^\pi \frac{1}{\theta} - \cos \theta \, d\theta$$

$$\textcircled{3} \int 4e^t - \frac{1}{t^3} \, dt$$

$$\textcircled{4} \int_{-1}^1 x^2(x-3) \, dx$$

$$\textcircled{5} \int \frac{1}{\cos^2(3x)} \, dx$$

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$$\textcircled{1} \int 3\sqrt{x} - \frac{1}{1+x^2} dx = 2x^{3/2} - \arctan(x) + C$$

$$\textcircled{2} \int_1^\pi \frac{1}{\theta} - \cos \theta d\theta = (\ln |\pi| - \sin(\pi)) - (\ln |1| - \sin(1)) = \ln |\pi| + \sin(1)$$

since $F(x) = \ln |\theta| - \sin \theta$ is an antiderivative of the integrand and $\sin(\pi) = \ln |1| = 0$.

$$\textcircled{3} \int 4e^t - \frac{1}{t^3} dt = 4e^t + \frac{1}{2t^2} + C$$

$$\textcircled{4} \text{ Rewrite } \int_{-1}^1 x^3 - 3x^2 dx = \left(\frac{1^4}{4} - 1^3 \right) - \left(\frac{(-1)^4}{4} - (-1)^3 \right) = -2$$

since $F(x) = \frac{x^4}{4} - x^3$ is an antiderivative of the integrand.

$$\textcircled{5} \int \frac{1}{\cos^2(3x)} dx = \frac{1}{3} \tan(3x) + C. \text{ Here you can think backward:}$$

when you take a derivative, you have to multiply by 3, so in taking an integral, we divided by 3. And always check by taking the derivative!